

Biostatistics Primer for the Radiologist

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Keywords: data analysis, medical statistics, regression analysis, statistical education, statistical tests

DOI:10.2214/AJR.13.11657

Received August 1, 2013; accepted after revision September 20, 2013.

Presented in part at the 2013 annual meeting of the ARRS, Washington, DC.

Supported by grant R01-AA017497 from the National Institutes of Health.

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AJR 2014; 202:W365–W375

0361–803X/14/2024–W365

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OBJECTIVE. The purpose of this article is to review the most common data analysis methods encountered in radiology-based studies. Initially, description of variable types and their corresponding summary measures are provided; subsequent discussion focuses on comparison of these summary measures between groups, with a particular emphasis on regression analysis.

CONCLUSION. Knowledge of statistical applications is critical for radiologists to accurately evaluate the current literature and to conduct scientifically rigorous studies. Misapplication of statistical methods can lead to inappropriate conclusions and clinical recommendations.

Radiology has been defined by technologic innovations and advancements focused on disease diagnosis and therapeutic interventions.

Conduct and dissemination of research evaluating the utility of imaging modalities and their applications have been and will continue to be vital components for guiding clinical recommendations and developing practice guidelines. Knowledge of basic statistical concepts will help the practicing radiologist to critically evaluate the literature and to make informed clinical decisions—the tenets of evidence-based radiology [1, 2]. Similarly, the appropriate use of statistical methodology and interpretation thereof are essential for conducting scientifically rigorous studies. However, introduction to research methodology, in general, and statistics in particular, is limited over the course of radiology training [3]. The clinical burden of radiologists, limited nontechnical resources centered on radiology research [4], and the lack of willingness to learn greatly impede future attainment and continued development of these skills.

Our objective is to present an introduction to the most common data analysis techniques encountered in the radiology literature. First, we begin with a description of variable types and summary measures for these variables. Second, data analysis is introduced, with a focus on methodology for comparing summary measures between multiple groups. Third, regression analysis is discussed, with particular emphasis on interpretation of the models. Finally, we conclude with a discus-

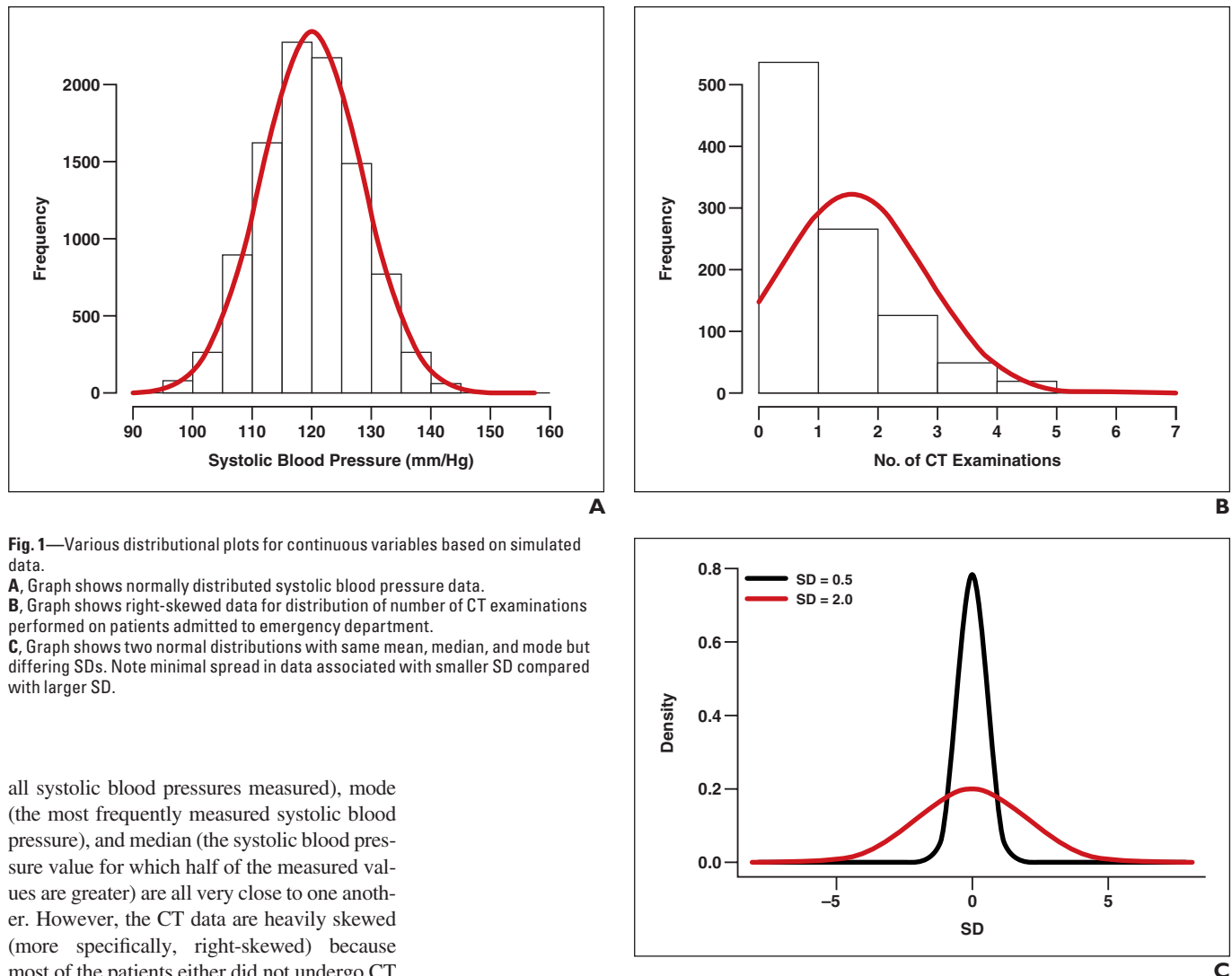
sion. We illustrate each topic with practical examples from radiology-related research. For interested readers, in-depth discussions of specific topics are referenced.

Data Types

Variables can be classified as either continuous or categorical and regarded as quantitative or qualitative measures, respectively. Continuous variables have equal intervals between values, such as height, weight, or the number of CT examinations performed for a patient during a specific hospital visit. In contrast, categorical (or discrete) variables can only take on predefined values, which are known as categories or levels. Categorical variables can further be classified as to whether they are nominal or ordinal in nature. Ordinal variables have a natural order, such as the increasing degrees of a pain scale. Nominal variables, such as sex, have no such intrinsic order. A binary (or dichotomous) variable is a categorical variable with only two values.

Summarizing Data

To summarize a variable in a meaningful way, we need to understand how the data are distributed. Figure 1 compares the difference between a distribution of systolic blood pressure measurements (Fig. 1A) and the distribution of the number of CT examinations performed in trauma patients (Fig. 1B), both of which are continuous variables. As shown, the systolic blood pressure data distribution is bell-shaped, which is also called a “normal” distribution. As a result, the mean (the average of



all systolic blood pressures measured), mode (the most frequently measured systolic blood pressure), and median (the systolic blood pressure value for which half of the measured values are greater) are all very close to one another. However, the CT data are heavily skewed (more specifically, right-skewed) because most of the patients either did not undergo CT or underwent just one during hospitalization. The mean, median, and mode differ in this instance. For nominal and ordinal variables, the summary measures will often be the mode and the median, respectively.

Along with the summary measure, the spread (variability or statistical dispersion) of each variable will be of interest. The SD is commonly used to describe the spread of a continuous variable. A smaller SD reflects less variability. Figure 1C illustrates this difference for two distributions in which the mean, median, and mode are similar; the SDs of the two differ and result in quite different distributions. Range, minimum, maximum, and quartiles are other useful descriptive measures commonly used for describing skewed data.

Graphic displays can also be useful methods for describing data. This visual presentation is as much an art form as anything else

and should be used judiciously to illustrate a concept that can otherwise not be clearly conveyed in the text of a manuscript. Figure 2 illustrates several of the most common forms of graphic summaries. Discussion and description of effective means of presentation can also be found elsewhere [5, 6].

Basics of Data Analysis

A comprehensive review of statistical techniques would require many textbooks. Therefore, we limit this discussion to techniques most relevant to radiologists [7]. Table 1 lists the portion of the radiology literature that is most accessible to radiologists familiar with specific statistical techniques and provides an orderly curriculum for radiologists seeking to enhance their knowledge of statistics and data analysis. Irrespective of

the type of analysis, results will be presented with three terms: point estimate, CI, and *p* value. Each of these terms is derived from aspects of hypothesis testing procedures (we refer interested readers to more detailed discussions of this subject [8–10]).

The point estimate is the comparison of the primary outcome variable between groups and the estimate of the true population difference on the basis of the sample. For example, in a study comparing gallbladder wall thickness measured on MRI between patients with and without cholangitis, the average (mean) difference of thickness between these groups will be the point estimate [11]. Similarly, we are interested in the reliability of this estimate; the CI, which is constructed on the basis of a confidence level, captures this imprecision. If we were to repeat a study many times, 95% of the

TABLE 1: Statistical Content by Category and Accessibility

Statistical Category	No. (%) of Articles Containing Methods (<i>n</i> = 669)	Cumulative Accessibility by Method (%)
None/descriptive	294 (44)	44
Basic Student <i>t</i> and <i>z</i> tests	157 (23)	52
Contingency tables: basic	106 (16)	60
Decision statistics: basic	98 (15)	67
Correlation/regression: basic	56 (8)	73
Nonparametric: basic	51 (8)	80
ANOVA: basic	37 (6)	83
Correlation/regression: nonparametric	29 (4)	85
Decision statistics: receiver operating characteristics	21 (3)	86
Transformation	16 (2)	87
Correlation/regression: advanced	16 (2)	88
Decision statistics: advanced	14 (2)	90
Contingency tables: advanced	13 (2)	91
Nonparametric: advanced	12 (2)	92
Survival analysis	12 (2)	94
Study design and analysis	8 (1)	96
ANOVA: advanced	5 (1)	97
Other	18 (3)	100

Note—ANOVA = analysis of variance. (Reprinted from [7])

time the point estimate obtained from any one study should fall within the range of the CI. Thus, the value obtained is not the probability that the single point estimate obtained falls within the CI, a subtle but important distinction. Finally, we would like to know whether the point estimate that we obtained represents a true difference—that is, whether it is statistically significant. The *p* value provides a measure of the likelihood of obtaining a value as extreme as the one observed, while assuming that in fact there was no difference. In most settings we set the significance level at *p* = 0.05 (any value less than 0.05, we will therefore consider a statistically significant difference), with a smaller *p* value providing statistical evidence supporting a difference between groups. Importantly, the *p* value is a measure of statistical significance rather than of clinical significance and is driven by sample size.

Statistical tests can be divided into two categories: parametric and nonparametric tests. The main assumption behind the use of parametric tests is the normal distribution of the primary continuous outcome. For example, it is reasonable to assume that the systolic blood pressure measurements have a normal (Fig. 1A) distribution in the population. We also

know that a blood pressure of 100 mm Hg is exactly twice as high as a pressure of 50 mm Hg. However, especially in radiology research, the outcome of interest often has a skewed distribution (Fig. 1B), making parametric tests inappropriate.

Nonparametric tests are also appropriate for analyzing noncontinuous data, such as a 5-point pain score, in which we know that a pain score of 4 is not twice as bad as a pain score of 2—we only know that it is worse than a pain score of 2. Nonparametric tests can also be used to analyze parametric data but may require slightly greater sample sizes to achieve the same statistical power. We have reviewed the most common analytical approaches in radiology research.

Analysis of Continuous Outcomes

Figure 3 summarizes different approaches for the analysis of continuous variables; a discussion of the common approaches follows.

Scenario 1: Student *t* Test (Two Groups)

For large samples, the Student *t* test can be used for comparing means between two independent groups. If the same subjects are tested twice, such as measuring the size of a hepatocellular carcinoma in patients be-

fore and after radiofrequency ablation [12], a paired Student *t* test is more suitable and provides considerably more statistical power for comparison of the mean difference.

Panzer and colleagues [13] used the Student *t* test to compare morphologic indicators of cam and pincer femoroacetabular impingement measured on multiplanar CT examinations between patients with and without herniation pits and found a statistically significant larger α angle in patients with herniation pits (55.23° vs 49.76°).

Scenario 2: Analysis of Variance (More Than Two Groups)

When comparison of means among more than two groups is of interest, analysis of variance (ANOVA) is most often used. The results of ANOVA will only show whether there is a difference between any two groups and is not a test of which specific groups differ—that is, this test determines whether all of the group means are equal. Repeated measures ANOVA can be used for dependent comparisons, and interpretation follows in a similar manner. To determine which groups differ from one another, post hoc testing between groups is required. However, performing many statistical tests or multiple comparisons can result in significant findings through chance alone. In these instances, a corrected significance level should be used. A brief overview of this topic can be found in the article by Tello and Crewson [9], and Bender and Lange [14] provide a more complete discussion.

Shepherd and colleagues [15] compared the impact of tube current reduction on radiation dose for patients undergoing three CT-guided spinal injection procedures for pain: facet joint injections, selective nerve root injections, and epidural block injections. ANOVA was used to compare radiation doses between these groups, and a statistically significant reduced total dose for these procedures was found.

Scenario 3: Nonparametric Analysis

For comparing continuous variables between two groups when the outcome is not normally distributed or with small sample sizes, paired and unpaired comparisons can be made using the Wilcoxon signed rank test and the Mann-Whitney *U* test, respectively. Importantly, in these tests, the means between two groups are not being compared; rather, these tests compare medians between the groups.

Nonparametric ANOVA, commonly referred to as the “Friedman” and “Kruskal-Wallis” tests,

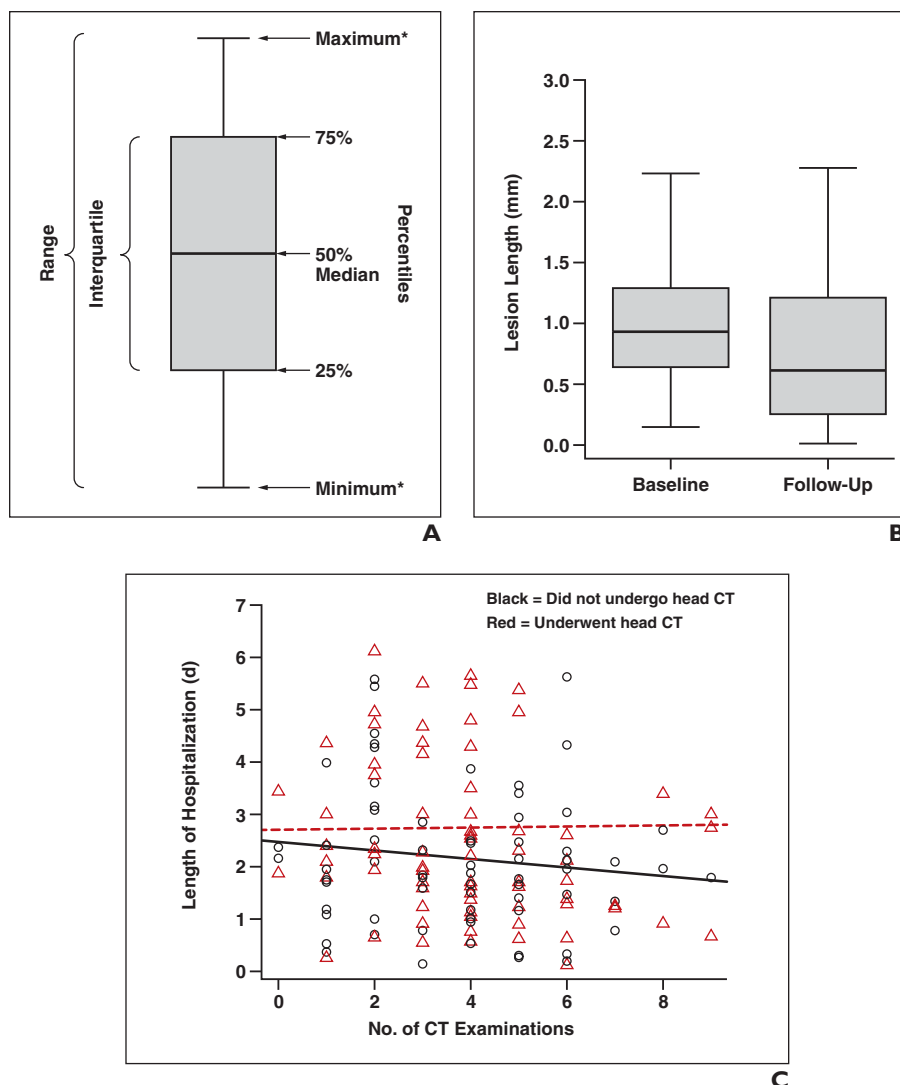


Fig. 2—Common approaches to graphical display of data.

A, Graph shows description of key components contained in boxplot. Shaded region represents 25th and 75th percentiles, and horizontal line therein represents median (50th percentile). Additionally, range of values is reflected with minimum and maximum values (*asterisks*). Whiskers (or fences) used in boxplots are implemented variably by different statistical software packages, which can lead to considerable confusion to their meaning [55, 56]. Different implementations are due to different exploratory rules of thumb. Intent is to indicate thresholds for data—values outside these whiskers are considered extremes or outliers. Because of this variability, it is appropriate to describe convention being used for whiskers and outliers in plot legend.

B, Boxplot compares tumor lesion length at baseline and after treatment on basis of simulated dataset.

C, Scatterplot shows number of CT examinations performed for each patient versus length of observation on basis of simulated data. Separate linear trend lines were fitted for groups of individuals who underwent head CT examination and those who did not. Note that there are few individuals with observed values at each end of x-axis. In this instance, these observations will contribute minimally to explaining variability of data.

can also be used for paired and unpaired measurements, respectively, for more than two group comparisons [16]. Inference for small sample sizes is conducted in a similar fashion to large sample methods because each is testing whether there is no difference between groups.

In a study of 64-MDCT and radiation exposure, Geyer et al. [17] compared filtered

back projection and adaptive statistical iterative reconstruction (ASIR, GE Healthcare) protocols in patients with cervical spine trauma. As part of this investigation, image quality between protocols was evaluated using two regions of interest at cervical vertebra 3 and 7 and was compared using the Mann-Whitney *U* test. Increased image noise was found when ASIR was used at both sites.

Scenario 4: Correlation

To this point, we have focused on comparisons of continuous and categorical outcomes between two or more groups. Correlation is used to measure the strength and direction of a linear association between a continuous exposure and a continuous outcome. Correlation is measured on a scale between -1 and 1 , with 0 representing no linear association, and -1 and 1 representing perfect negative and positive linear associations, respectively. One of the main limitations of the correlation is the inability to measure nonlinear associations. Figure 4 presents illustrative examples of correlation. Similarly, Figure 5 presents the Anscombe quartet [18], which consists of four very different datasets for which the correlation coefficient (0.82) is the same as well as the same fitted linear line yet the interpretation of the data differs for each.

To compare prostate tumor volume, a continuous variable measured using a multiparametric 3-T endorectal coil MR protocol with histopathologic correlation after prostatectomy, Turkbey et al. [19] used the Pearson correlation coefficient and found a positive correlation (0.633) between the two measures.

For an outcome variable that is normally distributed, the Pearson correlation coefficient is used, as was the case in the prostate tumor volume example. The Spearman rank correlation can be used for nonnormally distributed outcomes [4]. This test was used by Mandeville and colleagues [20] in a study of 35 patients with operable non-small cell lung cancer to compare the flow-extraction product and blood volume between dynamic contrast-enhanced CT parameters with the immunohistochemical markers of hypoxia, pimonidazole and glucose transporter 1.

Analysis of Categorical Outcomes

Scenario 5 is chi-square analysis. We often want to learn if there is a statistically significant association between two categorical variables. We can do this with a contingency table, which uses tests, such as the chi-square or Fisher exact tests, to assess the likelihood of this association. Analysis of categorical data is most often conducted with the chi-square test to determine whether the distribution of proportions is equal or different, with the analytical objective of comparing the observed number of observations with an expected number of observations. The chi-square test is so named because the chi-square distribution is used to determine the likelihood of the observed distributions [4]. Many generalizations

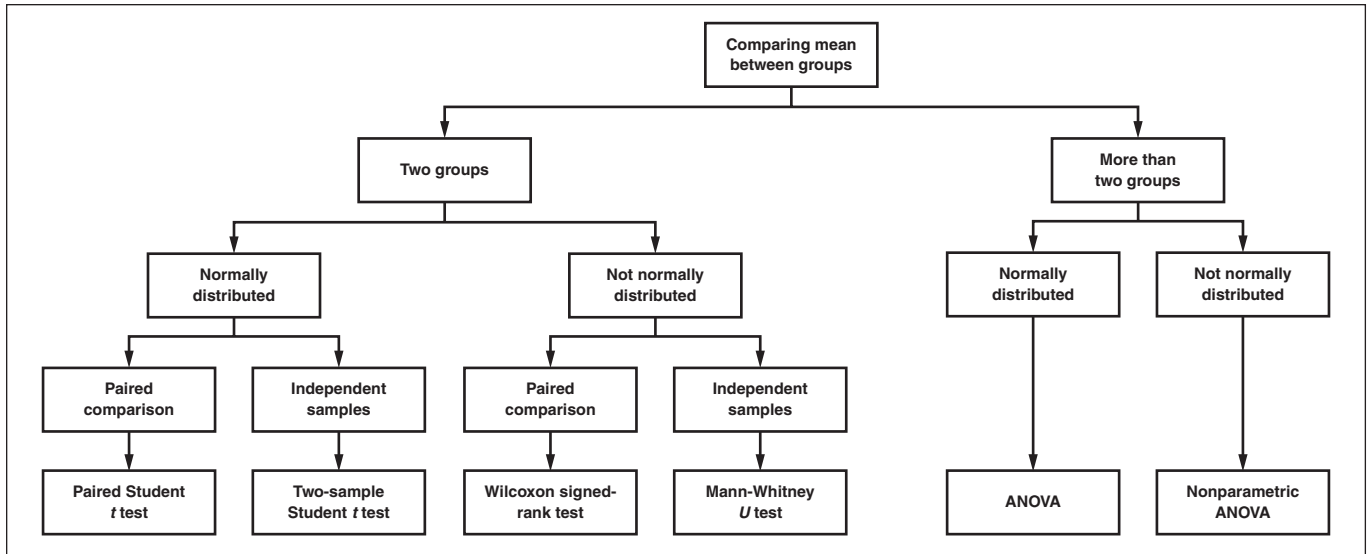
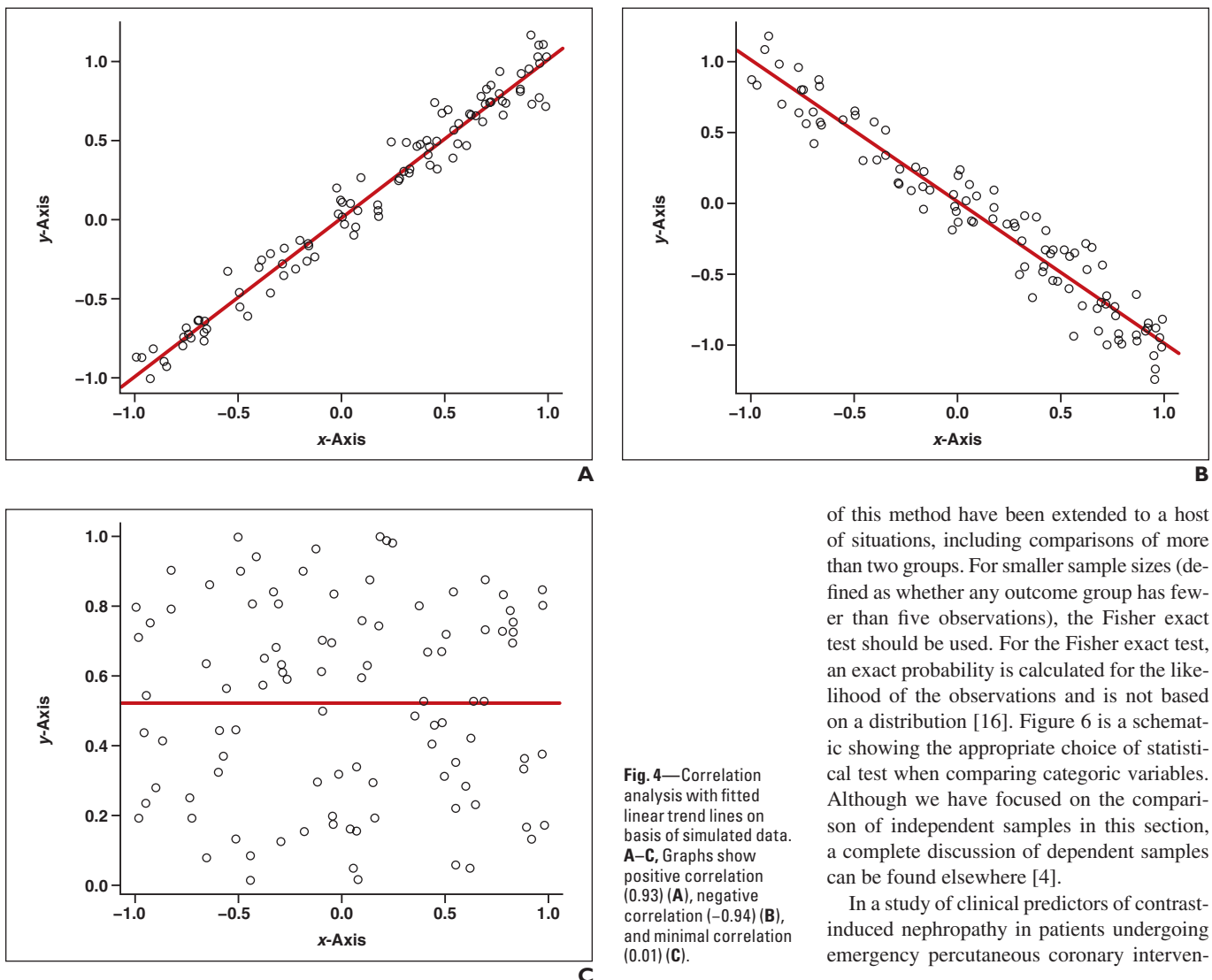


Fig. 3—Flowchart shows appropriate choice of statistical test when comparing continuous variables between groups. ANOVA = analysis of variance.



of this method have been extended to a host of situations, including comparisons of more than two groups. For smaller sample sizes (defined as whether any outcome group has fewer than five observations), the Fisher exact test should be used. For the Fisher exact test, an exact probability is calculated for the likelihood of the observations and is not based on a distribution [16]. Figure 6 is a schematic showing the appropriate choice of statistical test when comparing categorical variables. Although we have focused on the comparison of independent samples in this section, a complete discussion of dependent samples can be found elsewhere [4].

In a study of clinical predictors of contrast-induced nephropathy in patients undergoing emergency percutaneous coronary interven-

tion for acute coronary syndrome, Senoo and colleagues [21] used the chi-square test and reported a statistically significant difference in the proportion of men with contrast-induced nephropathy compared with those without (66% vs 79%).

Comparing Imaging Techniques

Scenario 6: Diagnostic Accuracy

Much of the methodology for comparison of imaging tests derives from the measurement error literature [22], on which a complete discussion of medical applications was presented by Pepe [23]. Sensitivity and specificity are the most commonly used statistical measures for comparing accuracy of a new imaging modality with a reference standard and to other imaging examinations. Sensitivity is the ability of a test to correctly identify an outcome among patients who have been diagnosed with that outcome using a reference standard approach. Specificity is the ability of a test to correctly classify healthy patients among those patients who were classified as nondiseased as a result of a reference standard test. Figure 7 shows how sensitivity and specificity are calculated via comparison with a reference standard. Other measures that are frequently discussed in the evaluation of accuracy of a new imaging modality are positive and negative predictive value (PPV and NPV, respectively). The PPV describes the proportion of positive tests that are actually positive on the basis of the currently available reference standard test; similarly, the NPV is the proportion of negative tests that are actually negative on the basis of the reference standard test.

A major problem with the PPV and NPV is that they are highly influenced by disease prevalence. For this reason, indexes such as the positive likelihood ratio and negative likelihood ratio are often preferred. The positive likelihood ratio is simply the sensitivity of the test divided by $1 - \text{specificity}$, and the negative likelihood ratio is calculated as $1 - \text{sensitivity} / \text{specificity}$ [24]. To clarify the relationship between these measures, the sensitivity and specificity can be thought of as pretesting measures (will the person's disease status be classified correctly), whereas PPV and NPV can be considered posttesting measures (was the person's disease status correctly classified). The actual significance of the test result can be communicated by the positive and negative likelihood ratios. These tests describe the probability of a positive test result for a person having a disease (or a neg-

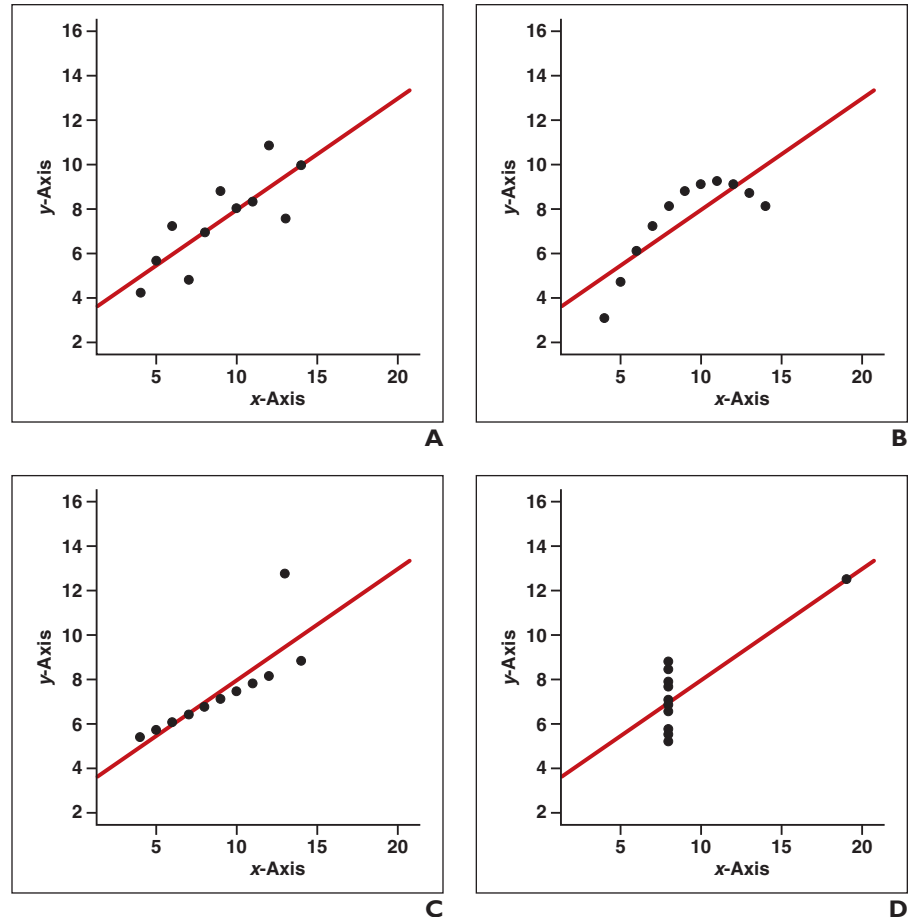


Fig. 5—Graphic display of Anscombe quartet [18].

A–D, Four datasets have similar statistical properties in that mean and variance of x and y variables are exactly same. Further, correlation (0.82) and linear regression line, $y = 3.0 + 0.5x$, are identical for each. Although statistical properties are similar, accurate interpretation would not be possible without data exploration.

ative test result for a person having disease) compared with the probability that a positive test result would be obtained in a person without disease (or a negative test result obtained in a person without disease). Importantly, the posttest odds of disease can be calculated by multiplying the likelihood ratio by the pretest odds of disease.

Rajaram et al. [25] compared the diagnostic performance of MR angiography (MRA) with pulmonary CT angiography for the diagnosis of pulmonary embolism and reported a high level of performance using MRA. Results from the study are presented in Figure 8 along with a description of how each of the summary measures that were reported were derived.

Scenario 7: Receiver Operating Characteristic Curve

Extending the applications of sensitivity and specificity, receiver operating character-

istic (ROC) curves are often used to compare the accuracy of diagnostic techniques [26, 27] and to account for variability between and among readers (who are generally radiologists in the imaging literature). This variability is often attributable to the individual subjectivity of the reader [28]. The ROC curve plots the sensitivity versus $1 - \text{specificity}$, or the true-positive rate versus the false-positive rate. In this manner, the subjective nature of the readers can be considered when comparing diagnostic techniques [29]. Evaluation of the ROC curve is performed by calculating the area under the curve (AUC), with higher values indicating better diagnostic accuracy [29]. Thus, these tests can be directly compared on the basis of their AUCs.

Rafferty and colleagues [30] compared the diagnostic accuracy for breast abnormalities between digital mammography in combination with breast tomosynthesis versus digital

mammography alone. These authors used an ROC curve to show that the addition of tomosynthesis resulted in more accurate diagnosis.

Scenario 8: Reader Agreement

Reader agreement studies [31]—how interpretations differ between and among two or more readers (interobserver) or how interpretations performed by the same reader at separate time points differ (intraobserver)—are used to ensure internal validity in studies using more than one reader. In comparing variability between continuous measurements, the Bland-Altman analysis is commonly used [32], whereas Cohen kappa is used for categorical variables. The Bland-Altman analysis examines the agreement between measurements and not the strength as is the case with correlation analysis. This is of interest for the reproducibility of methods and within- or between-reader comparisons, and the results are typically reported with the bias (i.e., the mean difference) of the measurements and limits of agreement or the expected range of values that could be expected for a measurement.

Returning to the comparison of categorical variables for reader agreement studies, an important consideration is that chance alone could potentially explain any observed association. The Cohen kappa statistic is a measure of reader agreement that accounts for chance alone; interpretation of this measure is described by Landis and Koch [33]. In the preceding discussion, although we have focused on the agreement between two readers, extensions to multiple readers can be performed using the generalized kappa statistic, and, accordingly, a weighted kappa statistic can be used when more than two categories of classification are possible.

Timmers et al. [34] compared the effectiveness of a training program for breast cancer assessment based on BI-RADS. Using the Cohen kappa statistic, the authors observed a significantly increased agreement between new screening radiologists and the expert panel of readers after implementation of the program.

Multivariable Analyses

In the previously introduced data analysis methods, one thing was common: We evaluated the association between one exposure and one outcome, while not considering other variables that could potentially influence such associations. For example, in a hypothetical study evaluating the association between mechanism of injury and CT use in trauma

patients, patient age should be considered because older patients tend to have more comorbidities; a resulting higher CT use could be influenced by these comorbidities. Multivariable regression analyses enable us to account for these covariates. Regression model selection and approach are dependent on the outcome variable type. For continuous outcomes, linear regression, Poisson regression, and negative binomial regression are more applicable to radiology research. For a categorical outcome, logistic regression and its subtypes are used.

As mentioned, multivariable analyses include other variables in the model in addition to the main exposure and outcome variables. As a result, the final point estimate and CIs reflect the association between the exposure and outcome after removing the potential effect of these covariates. There are two different approaches to determine inclusion of covariates in models. In an “a priori approach,” researchers use the existing literature to identify variables that could potentially influence the association between the exposure and outcome. Covariate selection is therefore evidence based. In “step-wise regression,” a data-driven approach for variable selection, the researcher begins with a set of variables that have been collected as part of the study and inclusion of covariates in the final model is determined by statistical significance, overall model fit, or a combination of the two. Fundamental to interpretation, multivariable analyses will only include subjects with data recorded for all variables—that is, the final analysis is a random subsample of all patients. Therefore, those individuals who have missing covariates will not be included, which could potentially introduce bias if the omitted subjects differ from those included [35].

Two important aspects of interpreting multivariable models are important. First, it is essential to understand that observed associations do not necessarily imply causation. Second, statistical significance of variables does not imply clinical significance.

Scenario 9: Linear Regression

The most commonly encountered multivariable analysis for the evaluation of the association between an exposure (continuous or categorical) and a continuous outcome is linear regression [36]. As reflected in the name of this analytical approach, the underlying assumption behind the use of this model is a linear relationship between the exposure and the outcome of interest. We will thus be concerned with finding the form of the best fitting linear line and not the strength as is the case in correlation. Any other relationship pattern—quadratic, U-shape, sigmoid, and so on—is not well evaluated using this approach (Fig. 6B). Results of linear regression will reflect the slope of the fitted line or the way in which a one unit change in the predictor variable influences the outcome.

Miao et al. [37] used linear regression to evaluate whether pericardial fat volume was associated with atherosclerotic calcification of the coronary arteries. The authors reported a statistically significant 0.42 increase of coronary artery plaque eccentricity with an increase of one SD of pericardial fat volume in men after adjusting for age, body type, medical history, plasma lipid levels, carotid intima-to-media thickness, and coronary artery calcification among others as potential factors that could influence the association between pericardial fat and atherosclerosis.

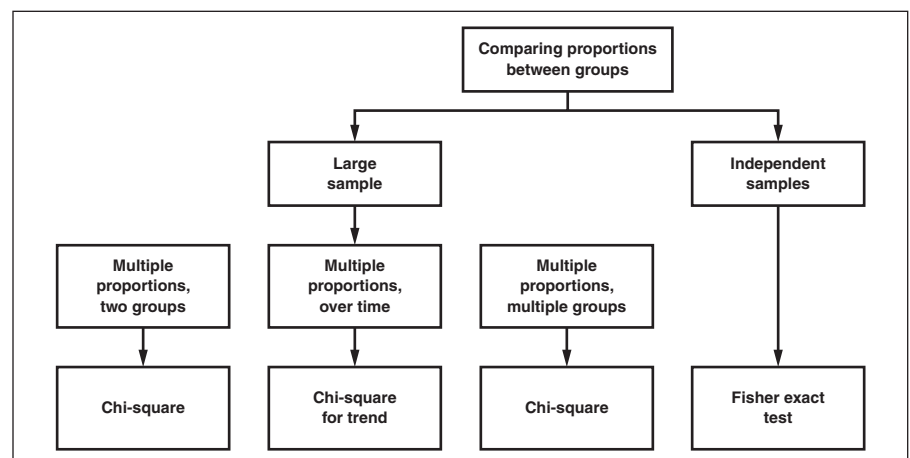


Fig. 6—Flowchart shows appropriate choice of statistical test when comparing categorical variables between groups.

Scenario 10: Logistic Regression

When only two categorical variables are evaluated, chi-square analysis is usually performed. When a dichotomous outcome and more than one covariate are considered, logistic regression [38] is the most commonly used multivariable analytical approach, which allows one to simultaneously consider the effects of multiple variables, including mixtures of categorical and continuous variables. Under this scenario, the association between a binary outcome and multiple continuous or categorical exposure variables is evaluated. The point estimate from logistic regression is called the "odds ratio" (OR), which is the ratio of the odds of an outcome in the exposed group to the odds of the outcome in the nonexposed group. The difference between probability and odds is that:

$$\text{odds} = \frac{\text{Probability}}{1 - \text{Probability}};$$

note that the logistic model infers a linear relationship between the predictor of interest on the log (odds) scale.

For a commonly occurring binary outcome (an outcome that occurs or is observed in > 10% of the study population), log-binomial regression is the preferred analytic approach. This stems from the fact that the OR will not accurately reflect the relative risk in this scenario [39]. Extending the idea of logistic regression, multinomial and ordinal regression can be used for evaluation of nominal and ordinal outcome variables with more than two categories, respectively.

Jung et al. [40] were interested in identifying patient subgroups at higher risk for allergic reactions to IV gadolinium contrast media. Using logistic regression analysis, they reported that women were at higher odds for adverse reactions compared with men, with an odds ratio of 1.69. In other words, the odds for an immediate hypersensitivity reaction to gadolinium-based MR contrast media was 69% higher for women compared with men.

Scenario 11: Poisson and Negative Binomial Regression

When the outcome is a continuous variable with skewed distribution, using linear regression could result in misleading results [41]. One solution is categorizing the contiguous outcome to two categories and to use a logistic regression model. The disadvantage of this approach is the loss of potentially valuable data in the process of categorizing the outcome. The more appropriate negative binomi-

al and Poisson regression analyses are two of the most commonly used methods for dealing with skewed continuous outcomes. Both of these models are considered parametric models, with Poisson analysis a special case of the negative binomial; this also assumes that the mean and variance are equal. Detailed discussion regarding the pros and cons of using linear, Poisson, and negative binomial regression for analysis of continuous outcome can be found elsewhere [41]. The point estimates in both the negative binomial and Poisson regressions are interpreted as incidence rate ratios or the percentage difference in outcome

for a one-unit change in the main exposure of interest. Many extensions of this model are available to accommodate different forms of count data [42].

To evaluate the trend in the use of CT for elderly trauma patients admitted to a level 1 trauma center for fall-related injuries from 1996 to 2006, Roudsari and colleagues [43] used negative binomial regression. The authors reported a 7% annual increase in use of head CT in 2006 compared with 1996 after adjustment for age, year of admission, sex, insurance status, ethnicity, ICU admission status, injury severity score, and final disposition.

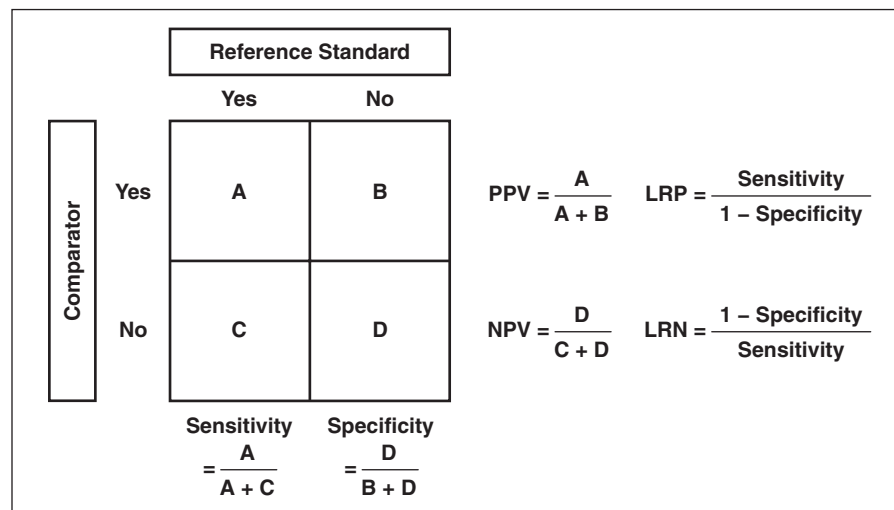


Fig. 7—Schematic shows data setup for calculating primary measures for evaluating diagnostic performance between two modalities using reference standard approach. PPV = positive predictive value; NPV = negative predictive value; LRP = likelihood ratio positive; LRN = likelihood ratio negative.

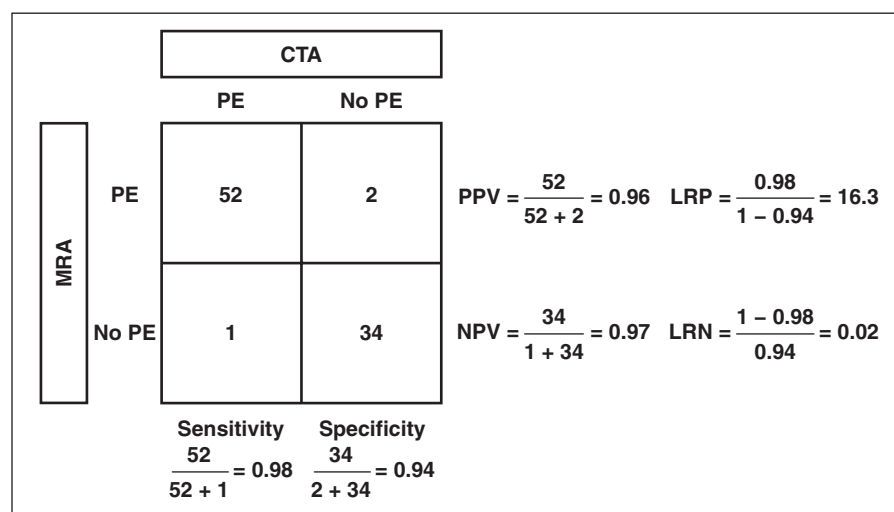


Fig. 8—Schematic shows calculation of diagnostic performance obtained by Rajaram et al. [25], comparing diagnostic performance of MR angiography (MRA) with pulmonary CT angiography (CTA) for pulmonary embolism (PE) diagnosis. PPV = positive predictive value; NPV = negative predictive value; LRP = likelihood ratio positive; LRN = likelihood ratio negative.

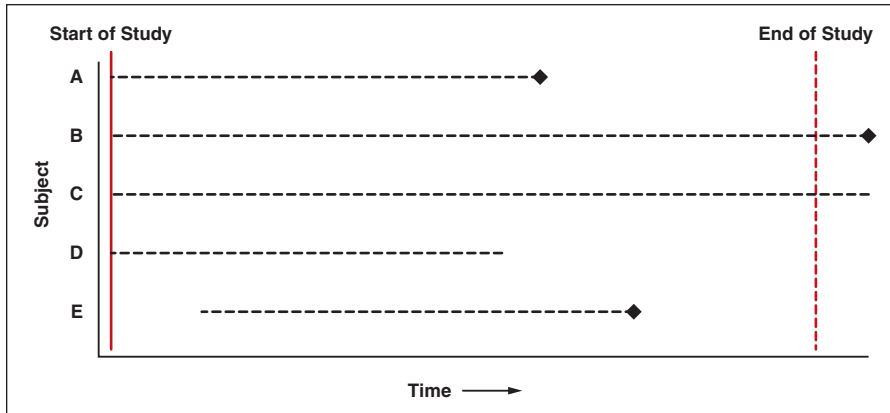


Fig. 9—Graph shows common censoring mechanisms encountered in time-to-event analysis. Note subject A developed outcome (♦) while under study. Subjects B and C were both right censored with outcome not observed while under observation. Subject B later went on to develop disease. Subject D was lost to follow-up and right censored. Subject E was left truncated because of recruitment into study after it had commenced; this subject later developed outcome.

Scenario 12: Survival Analysis

Survival analysis [44], or time-to-event analysis, is commonly used in the literature to evaluate the potential influence of a new imaging modality on patient outcome. For example, multiple studies have evaluated the influence of mammography on patient survival after breast cancer [45–47]. The outcome of interest for these analyses is a dichotomous variable—did the event occur: yes/no—with consideration of the time to occurrence. Thus, survival analysis can be conceptualized as an extension of logistic regression with the inclusion of time. Survival analysis enables simultaneous consideration of the effects of multiple variables on survival time. One complication of using survival time as a variable is that most studies contain censored data, in which one does not know the exact survival time of a subject. This can happen when a subject survives past the end of the study period or when a subject is lost to follow-up. Figure 9 provides examples of common censoring in studies.

Descriptive statistics for survival analysis will often present the median time to event and the Kaplan-Meier curve, which is a plot of the proportion of events occurring over time. Interest will often focus on whether specific variables are associated with the time to event. Unadjusted analysis of survival between two or more groups is usually performed using the log-rank test. For many clinical trials, this will be the primary analytic method. However, in cohort studies, adjusted (multivariable) analyses will commonly be performed with Cox proportional hazards regression [44]. The point estimate in this case is a hazard ratio, which is the ratio of the instantaneous risk of

outcome in the exposed group to the hazard or risk of outcome in the control group after adjustment for other covariates [44, 48].

To investigate whether the standardized uptake value (SUV) of ^{18}F -FDG on PET was a predictor of survival in patients with head and neck cancer, Torizuka and colleagues [49] used a time-to-event analysis and reported that a one-unit increase in SUV was associated with an 11% increased risk of mortality and a 10% increased risk of persistent or recurrent tumor.

Scenario 13: Trend Analysis

The secular trends (increasing, decreasing, or no change) in practice patterns are often of interest to resource planners, such as the case of data presented in Figure 10. Several analytical approaches are available to answer these questions. The chi-square test for trend is the

most basic analysis. In this unadjusted analysis, the scientific question being tested is whether there has been a linear trend in proportions over time. Similarly, a linear trend could also be investigated using multivariable linear regression to adjust for covariates. As explained earlier, assuming the outcome of interest has a linear distribution over time may be an untenable assumption. Likewise, Poisson or negative binomial regression could be used to evaluate the trend in occurrence of skewed continuous outcomes. In the previously described study by Roudsari et al. [43], regarding the trend in the use of CT for fall-related injuries in elderly trauma patients, the primary interest was in evaluating whether there was a change in CT use over the study period. The authors showed that on average the use of head CT increased by 7% in 2006 compared with 1996 using negative binomial analysis [43].

Multivariable time series models [50], while more difficult to construct and interpret, can also be used to analyze continuous or categorical outcomes. The primary strength of these methods is that the correlation of observations (dependence) can be accounted for, which is often not considered in other regression techniques. However, we note that mixed-model approaches [51, 52] are complicated regression models that explicitly account for the dependence of observations, such as the case of clustering. For example, when analyzing yearly CT use, years closer to one another will be more closely related than years farther apart. Interest will often include the identification of an underlying trend of the data (such as declining use) and whether there is a cyclic pattern, such as seasonal increases associated with increased number of patients. Finally, time series are also used to

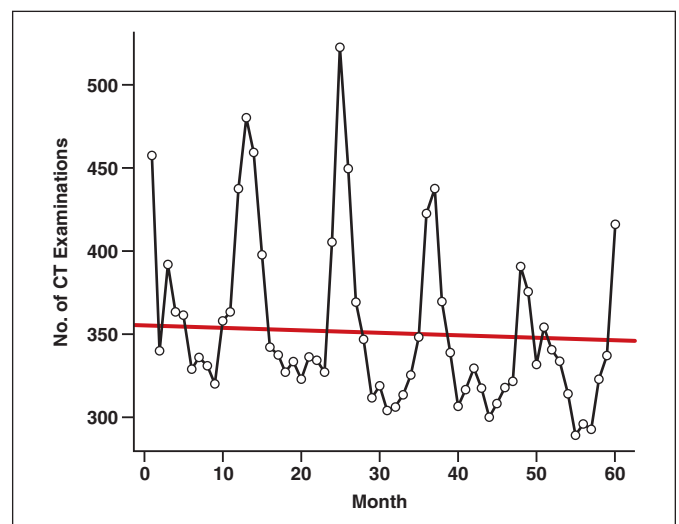


Fig. 10—Graph shows simulated single center 5-year monthly time series of number of CT examinations performed, with linear trend line fitted to data. Moderate secular decline in CT examinations performed is indicated; however, monthly variation due to differential number of patients in each month is masked.

forecast, or predict, future values. Siström and colleagues [53] used time series analysis to determine whether implementation of radiology order entry and decision support systems impacted the CT growth rate at an academic medical center and reported a decrease in CT volume (274 studies/quarter) and growth (2.75%/quarter) after implementation.

Conclusion

In this article, we have provided a brief overview of the fundamental aspects for describing, analyzing, and interpreting data. For the radiologist, these aspects are critical for evaluating and conducting research. Considering the ubiquity of advanced methodologies in the radiology literature, analytical competency is now more important than ever. Statistical consultations will likely be required when conducting clinical research; therefore, statistical literacy will be essential for collectively designing an appropriate analytical approach specific to the scientific question. To provide continued leadership and excellence in the specialty, radiologists should assume the responsibility of showing the technologic utility of imaging tests [54]; this onus is largely placed on well-trained academic radiologists. Lacking this, inappropriate conclusions and clinical recommendations will result from poorly conducted research.

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